COURSE NLP301 / MODULE NL

Modern NLP with Transformers

Learning goal: explain core concepts, compare related methods, and apply them to realistic machine learning or educational technology scenarios.

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01. What does self-attention do in a transformer?

For each token, self-attention combines value vectors from other allowed positions using normalized query-key similarity. Multiple heads can learn different interaction patterns; the attention mask determines visible context. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

02. Why is positional information necessary?

Attention alone does not encode token order. Positional encodings or learned position embeddings provide order information so phrases with the same words in different arrangements can differ. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

03. How do encoder and decoder models differ?

A bidirectional encoder represents tokens using surrounding input context and is useful for classification and extraction. A causal decoder predicts the next token based on prior tokens and is suited to text generation. A useful way to understand how do encoder and decoder models differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

04. What are BERT, GPT, and T5 used for?

BERT is primarily a pretrained encoder for understanding tasks; GPT denotes a family of causal decoder models for generation. T5 frames varied tasks as text-to-text using an encoder-decoder design. Actual capabilities depend on model version and training. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

05. What is fine-tuning?

Fine-tuning continues training a pretrained model on labeled task data to adapt its behavior. Use separate validation data and compare with prompting or retrieval baselines; monitor overfitting and licensing constraints. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

06. What is transfer learning in NLP?

Transfer learning reuses representations learned from large corpora for a narrower task. It reduces task-specific data needs, though domain shift and specialized vocabulary can still require adaptation and careful evaluation. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

07. How is Hugging Face Transformers commonly used?

Load a compatible tokenizer and pretrained model, tokenize inputs with padding and truncation, then run inference or fine-tune. Record model revision, maximum length, preprocessing settings, and evaluation data for reproducibility. In practice, it helps to think of this as a sequence of steps rather than a single operation. Each step affects the information available to the next stage, so implementation choices should be validated on representative data. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.